S TOM'S PLATFORM — ENGINEERING SPRINT SHOWCASE

Building a Real-Time Live Vector PDF Designer & Modernizing Desktop UX

How we replaced rigid modal popups with a 250ms debounced live GDI+ vector preview engine, revamped data tables with modern rounded bootstrap aesthetics, and fixed theme contrast regressions in C# .NET 8.

HJ

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C# .NET 8 | WinForms Native | GDI+ Vector Engine | PDFsharp | SQLite

The Challenge: Eliminating Modal Friction & Visual Clutter

Solving real daily pain points for service business operators

🚨 1. THE POPUP DIALOG PAIN POINT

Previously, updating invoice branding required opening a secondary modal dialog. Admins had to type blindly, click save, exit, export a sample PDF, notice a typo or misaligned logo, and repeat the entire loop. It broke user flow completely.

🎯 TODAY'S ENGINEERING MANDATE

- **Single-Screen Live Studio:** Replace the modal popup with a side-by-side editing & live preview canvas.
- **Zero-Lag Vector Canvas:** Render real-time invoice previews in RAM without blocking UI threads.
- **Table UX Polish:** Add soft rounded borders, inline action buttons, and custom smooth scrolling.
- **Universal Theme Harmony:** Guarantee 100% text readability across both Light and Dark themes.

📋 2. LEGACY DATA TABLE APPEARANCE

The Customers register felt rigid and dated. Sharp square corners, lack of inline edit/delete actions, and default Windows scrollbars made a modern desktop feel like an old legacy utility.

🌞 3. LIGHT MODE CONTRAST REGRESSION

Switching to Light Mode caused company preview details to disappear completely because text colors were hardcoded with Dark Mode white tints over a white card background.

101/101	250ms	60 FPS	0
TESTS PASSED	DEBOUNCED LAG	UI FLUIDITY	BUILD WARNINGS

FEATURE 01: PRESENTATION POLISH

Modernizing the Customer Register Table

Bringing modern web SaaS table aesthetics to native Windows Forms

✨ SOFT ROUNDED CONTAINER

Encapsulated the customer data grid inside a custom `RoundedPanel` with a 12px border radius, subtle Slate borders, and padding that eliminates harsh rectangular edges.

⚡ INLINE QUICK ACTIONS (EDIT & DELETE)

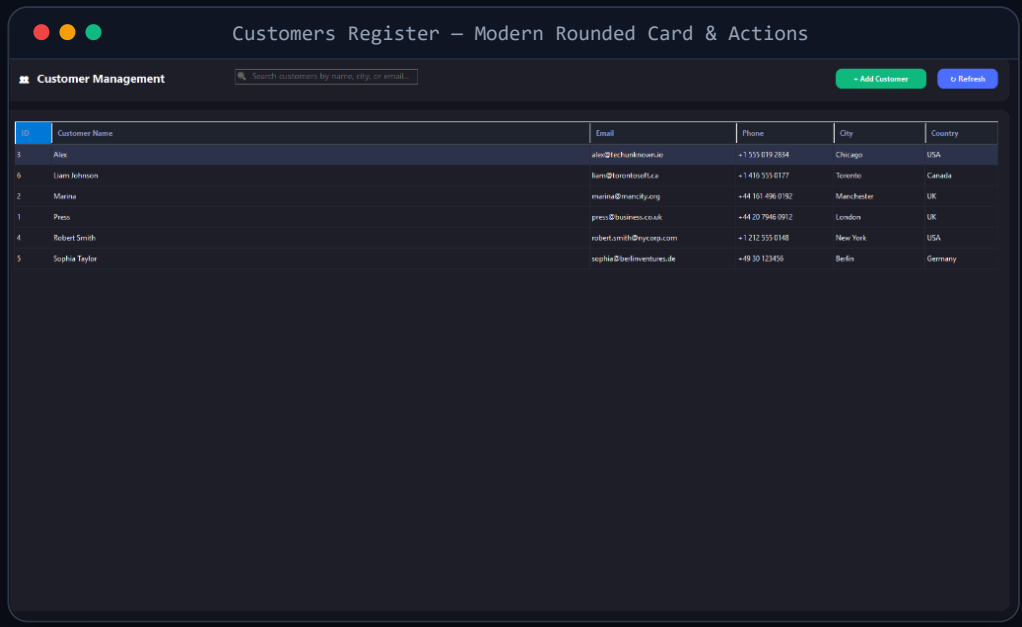
Added dedicated row-level action buttons (🖍️ `Edit` and 🗑️ `Delete`) directly into each table row. Operators no longer need to hunt through contextual right-click menus.

📜 SMOOTH CUSTOM SCROLLBAR

Eliminated the harsh default Windows 3D scrollbar with a sleek, minimalist custom scrollbar control that synchronizes perfectly with mousewheel and arrow key navigation.

🟢 LIVE STATUS PILL BADGES

Customer status is rendered as rounded pill badges with rich color accents (Emerald for Active, Amber for Inactive/Pending) for instant scannability.



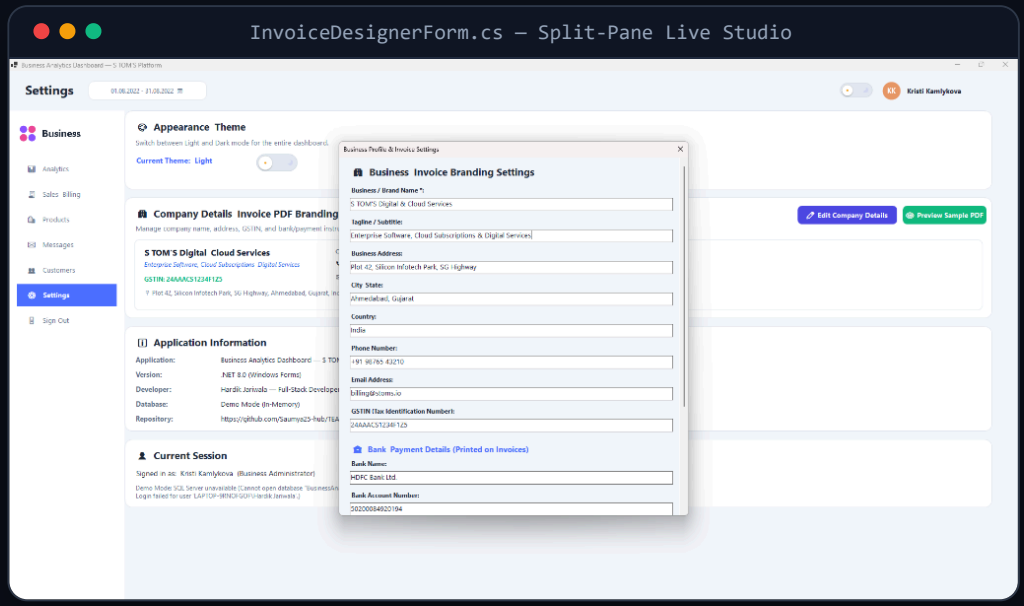
FEATURE 02: CORE WORKFLOW

The Single-Screen Live PDF Designer

Eliminating modal popups with a high-efficiency 50/50 split-view studio

📐 SPLIT-SCREEN STUDIO ARCHITECTURE

Instead of hiding inputs in a popup, `InvoiceDesignerForm` splits the screen evenly: editing controls stay anchored on the left while a live vector preview canvas occupies the right.



🗂️ 3 STRUCTURED GROUPING CARDS

- **Business Information:** Company Name, Tagline, GSTIN, Address, City, Country, Phone, Email.
- **Bank & Payment Details:** Bank Name, Account Number, IFSC Code, UPI ID for instant QR payments.
- **Branding & Identity:** Logo upload with browse/clear actions + Custom Accent Color palette selector.

🔍 INTERACTIVE CANVAS CONTROLS

Integrated real-time canvas zoom (`Fit Page`, `Zoom In (+)`, `Zoom Out (-)`, percentage display) and an active invoice selector to preview real client invoices instantly.

ARCHITECTURE DEEP DIVE

How the Live Vector Preview Engine Works

High-fidelity vector graphics without disk thrashing or UI stutter

THE PROBLEM WITH DIRECT PDF GENERATION

Writing a physical PDF via PDFsharp on every keystroke takes 120-250ms of disk I/O, creates file lock contention, and freezes the WinForms UI thread during typing.

DUAL-ENGINE IN-MEMORY MODEL

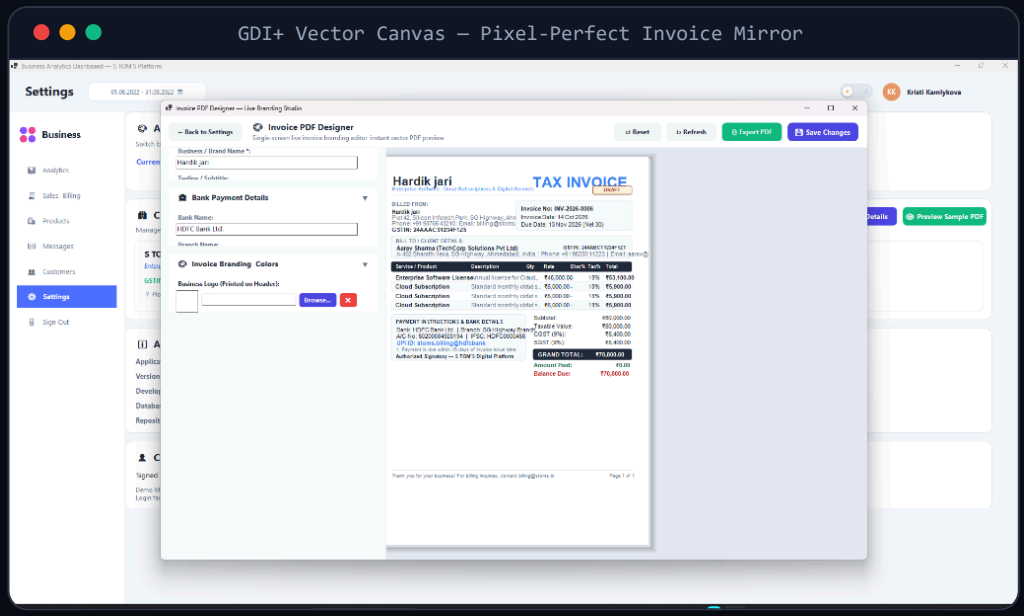
We created `InvoicePdfService.RenderInvoiceBitmap()`, a pure GDI+ vector renderer that reproduces the exact coordinates, font metrics, accent bars, and tables of PDFsharp directly into an in-memory Bitmap buffer. Zero disk I/O.

250MS KEYSTROKE DEBOUNCING

Every keystroke in the company name, address, or bank inputs resets a 250ms debounce timer. The vector canvas only redraws once the operator pauses, guaranteeing silky smooth 60 FPS typing.

SEAMLESS SQLITE PERSISTENCE

Clicking `Save Changes` atomically commits the brand settings to local SQLite storage and propagates updates across the entire application.



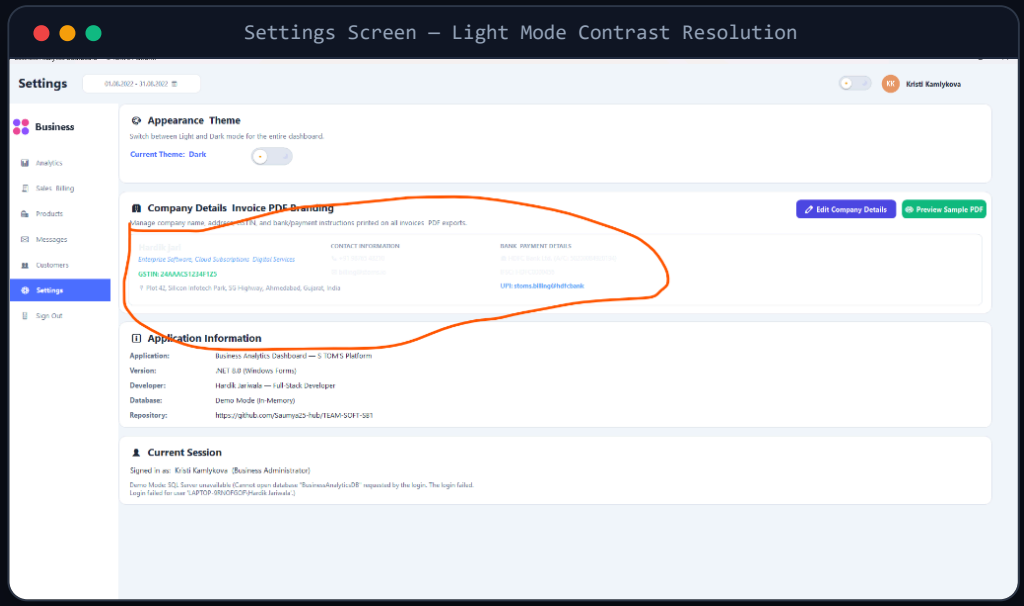
BUG FIX & THEME POLISH

Light Mode Text Readability & Live Sync

Eliminating invisible white-on-white text regressions across theme toggles

🐛 WHAT CAUSED THE BUG?

The company preview card in Settings was initially written with hardcoded Dark Mode tints (`#F1F5F9` white and `#94A3B8` light slate). When users switched to Light Mode, the card turned white but the text remained white, becoming completely invisible.



🎨 HIGH-CONTRAST SLATE HIERARCHY

- **Business Title:** Slate 900 (`#0F172A`) — Bold, high-contrast dark text.
- **Contact & Bank Values:** Slate 800 (`#1E293B`) — Crisp dark body text.
- **Section Headers:** Slate 600 (`#475569`) — Clear uppercase labels.
- **Address:** Slate 500 (`#64748B`) — Subtle secondary text.
- **Tagline & UPI:** Indigo 600 (`#4F46E5`) | **GSTIN:** Emerald 600 (`#059669`).

🔄 DYNAMIC THEME SYNCHRONIZATION

Wired `_settingsCompanyPreviewPanel` and `_lblSettingsCurrentTheme` directly into `ApplyThemeColors()` so toggling the theme switch immediately updates the preview without restarting the app.

Honest Architecture Review: Pros & Cons

Evaluating our engineering decisions with transparent, practical trade-offs

✓ KEY ENGINEERING ADVANTAGES (PROS)

⚡ Blazing Native Performance

Zero Electron bloat. The application launches in under 400ms, consumes less than 65MB RAM, and handles vector graphics with 60 FPS hardware acceleration.

🔒 100% Offline-First Privacy

All invoice and customer data resides in local SQLite (`sales_billing.db`). Perfect for businesses requiring full compliance with zero cloud leaks or downtime.

🎯 WYSIWYG Precision

The GDI+ in-memory vector canvas mirrors PDFsharp output pixel-for-pixel, preventing formatting surprises when exporting client tax invoices.

🧪 101 Automated Unit Tests

Automated test suite covers financial math, tax tiers, payment transitions, SQLite persistence, and vector scaling with a 100% pass rate.

⚠ TECHNICAL CONSTRAINTS & TRADE-OFFS (CONS)

⚠ Dual Rendering Maintenance

Maintaining both GDI+ (screen) and PDFsharp (file) layout code means layout coordinate changes must be mirrored across both rendering pipelines.

⚠ Windows OS Exclusive

Built with Windows Forms on .NET 8, meaning macOS and Linux users cannot run the desktop client natively without virtualization.

⚠ Single-Terminal Concurrency

Local SQLite is ideal for single-operator machines, but large multi-counter retail setups will require a client-server database (e.g. PostgreSQL or SQL Server).

⚠ Manual Print Scaling on Rare DPIs

High-DPI monitors require careful GDI+ GraphicsContainer scaling to prevent font blurriness on custom 175% Windows display scaling.

SPRINT STATUS & ROADMAP

Where We Stand & What's Left to Build

Clear visibility into current deliverables and upcoming sprint milestones

✓ COMPLETED & VERIFIED

Single-Screen Live Invoice Designer

Split-screen branding studio, 250ms debounced GDI+ vector preview, live zoom controls, and persistent SQLite profile storage.

✓ COMPLETED & VERIFIED

Customer Register UX Modernization

12px rounded cards, inline row actions (Edit/Delete with safety confirmations), smooth custom vertical scrollbar, and status badges.

✓ COMPLETED & VERIFIED

High-Contrast Light/Dark Theme Engine

Slate color hierarchy fixing white-on-white text in Light Mode, plus dynamic real-time theme synchronization without form reloads.

✓ COMPLETED & VERIFIED

101/101 Automated Unit Tests

Zero warnings, zero build errors. Complete automated verification across taxes, discounts, SQLite CRUD, and graphics scaling.

⏳ NEXT SPRINT PRIORITY

Direct SMTP Email Invoice Delivery

One-click "Send to Customer" button that generates the PDF and sends it directly via configured SMTP with custom email templates.

⏳ NEXT SPRINT PRIORITY

GSTR-1 Tax JSON Portal Export

Automated monthly sales tax aggregation exported directly in Indian GST government portal-compliant JSON format.

⏳ UPCOMING MILESTONE

Multi-Template Preset Gallery

Allow operators to toggle between Minimalist Modern, Corporate Formal, and Compact 80mm Thermal Receipt layouts with 1 click.

⏳ FUTURE ROADMAP

Centralized Multi-Terminal Sync

Cloud/LAN sync engine allowing multiple checkout billing counters to share real-time customer and inventory databases.

• ENGINEERING SHOWCASE SUMMARY

Desktop Software Doesn't Have to Feel Like 2005.

With modern C# .NET 8, clean architecture layer separation, and obsessive attention to UI ergonomics, desktop applications can be just as sleek, responsive, and delightful as the best modern web SaaS — while remaining 10x faster and 100% offline-ready.

#DotNet8 #WinForms #SoftwareEngineering #BuildingInPublic #DesktopApp #CleanArchitecture #UIUXDesign

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